From **Normal** to **Predictive**: The **RequisitionIQ** Framework

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The **RequisitionIQ** Framework

Level 4: Advanced Intelligence
Learn from requisition reality

Level 3: Pipeline Health
You can’t manage requisitions when you can’t monitor

Level 2: Context Intelligence
Predict the outcome of requisitions

Level 1: Historic Context
Define normal filled requisitions

- Sampling
- Context
- Signals
- Z-Scores
- Survival
- Target Goals

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The **RequisitionIQ** Framework

Level 4: Advanced Intelligence
Learn from requisition reality

- *Supervised Learning*
- *Unsupervised Learning*
- *Deep Learning*

Level 3: Pipeline Health
You can't manage requisitions when you can't monitor

- *Decision Analytics*
- *Classification*
- *Clustering*
- *Big Data Analytics*

I will only build what I learn in class…

Level 2: Context Intelligence
Predict the outcome of requisitions

- *Advanced Statistics*

Level 1: Historic Context
Define normal filled requisitions

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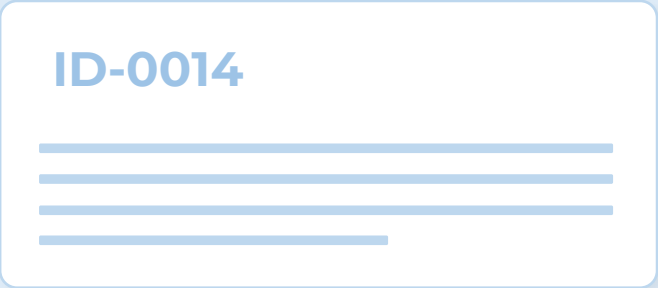
The **RequisitionIQ** Framework

1 Unit of Analysis



Unit: one *filled* requisition
Sample Size: 40 filled requisitions
Why 40: ≈ 6 months of recruiter output
Purpose: balance statistical stability with operational relevance

Sampling — Context — Signals

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The **RequisitionIQ** Framework

1 Unit of Analysis

ID-0014

Unit: one *filled* requisition
Comparison set: 40 comparable fills
Why 40: ≈ 6 months of recruiter output
Purpose: balance statistical stability with operational relevance

Sampling — Context — Signals

RequisitionIQ is NOT…

Data Quality Management
All data entering the systems has already been scrubbed, validated, and free from NaN and Nulls from a separate DQ process.

Predictive or Prescriptive Analytics
ReqIQ v1.0 produces data sets with descriptive signals only.

Individual Performance Scoring
Recruiters are used as contextual lenses, not as the unit of evaluation.

A Dashboard or BI Tool
ReqIQ generates signals. Visualization is downstream.

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Quarterly Business Review

Talent Acquisition

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“I want to **compare Q3** to **Q2**”

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Without context, numbers are meaningless…

89	62%	15	62
applicants	reviewed	days to first screen	time to fill

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89
applicants

62%
reviewed

15
days to first screen

- is this **good?**
- is this **fast?**
- is this **normal?**

62
time to fill

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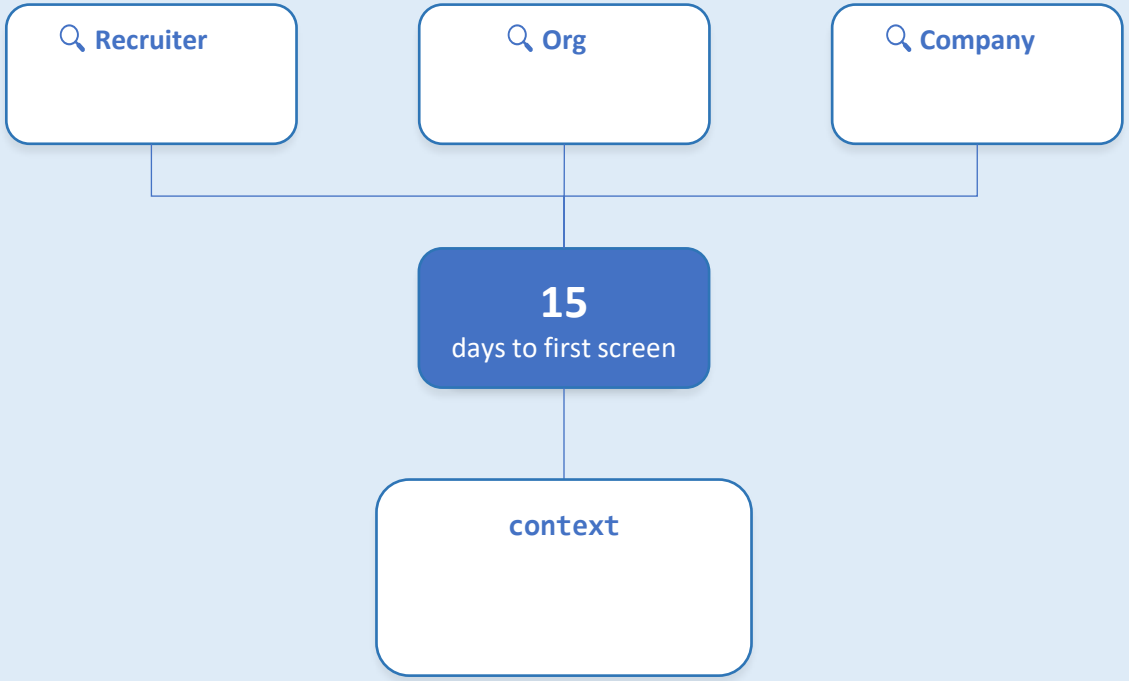
The **Context "Lens"**



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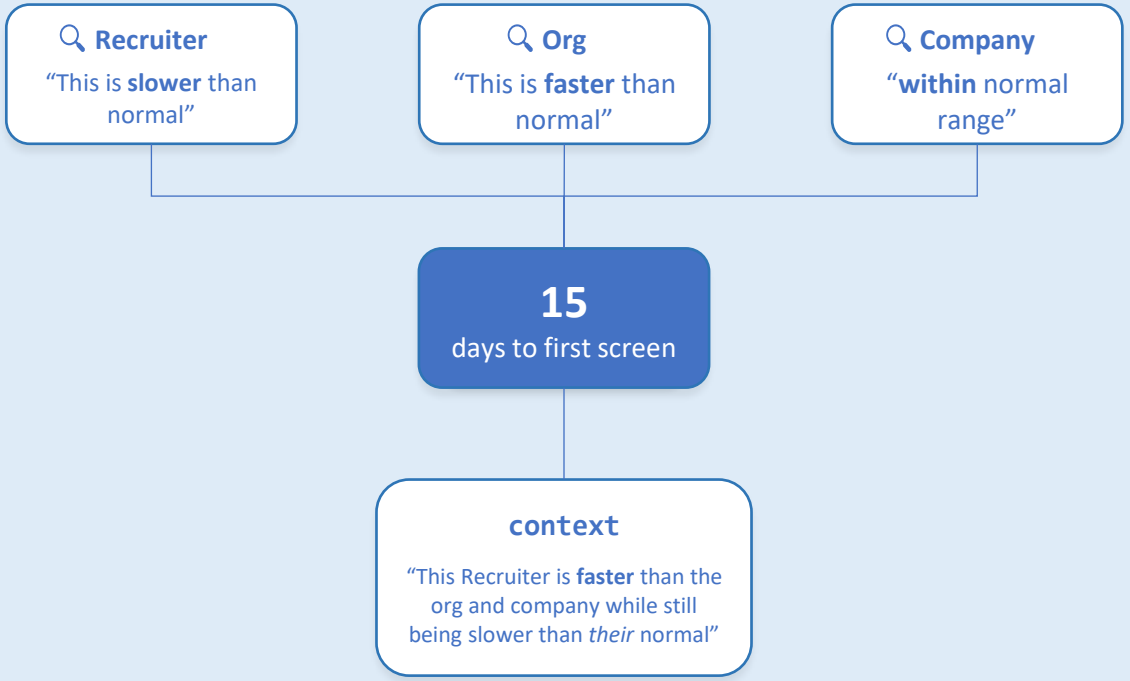
The **Context "Lens"**



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The **Context "Lens"**

89
applicants total

62%
reviewed in x days

15
days to first screen

62
time to fill

each metric gets a **fair** evaluation

Recruiter
"This is **slower** than normal"

Org
"This is **faster** than normal"

Company
"**within** normal range"

15
days to first screen

context
"This Recruiter is **faster** than the org and company while still being slower than *their* normal"

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The **Context "Lens"**

Recruiter

Org

Company

1 Requisition

ID-0014

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The **Context "Lens"**

1 Requisition

ID-0014

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

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The **Context "Lens"**

1 Requisition

ID-0014

	Recruiter	Org	Company
applicants total	context	context	context
reviewed in x days	context	context	context
days to first screen	context	context	context
time to fill	context	context	context

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The **Context "Lens"**

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

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The **Context "Lens"**

	Recruiter	Org	Company
days to first screen			

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The **Context "Lens"**

	Recruiter	Org	Company
days to first screen			

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The **Context "Lens"**

	Recruiter	Org	Company
days to first screen	Operational Reality		
	• recruiter • country • job_group • job_family • job_level	• country • job_group • job_family • job_level	• job_group • job_family • job_level

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1 Requisition

ID-0014	recruiter	country	job_group	job_family	job_level

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering	Entry

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering	Entry

I want to **compare Q3 to Q2**

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering	Entry

I want to **compare Q3 to Q2**

ID	Recruiter	Country	Job Group	Job Family	Job Level
2084	Christian	UK	Tech	Engineering	Entry
6184	Christian	UK	Tech	Engineering	Entry
6444	Christian	UK	Tech	Engineering	Entry
6644	Christian	UK	Tech	Engineering	Entry

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering	Entry

I want to **compare Q3 to Q2**

ID	Recruiter	Country	Job Group	Job Family	Job Level
2084	Christian			Engineering	Entry
6184	Christian			Engineering	Entry
6444	Christian			Engineering	Entry
6644	Christian			Engineering	Entry

n=4 → not enough

Minimum sample size: n=40
(≈6 months recruiter output)

** the observation (**ID-0014**) will never appear in the n=40 sample population

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering	Entry

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering	~~Entry~~

Drop a column a try again…

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1 Requisition

ID-0014	Christian	UK	Tech	Engineering

I want to **compare Q3 to Q2**

ID	Recruiter	Country	Job Group	Job Family
4837	Christian	UK	Tech	Engineering
2084	Christian	UK	Tech	Engineering
6184	Christian	UK	Tech	Engineering
6444	Christian	UK	Tech	Engineering
6644	C			Engineering
1925	C			Engineering
4765	C			Engineering
1355	Christian	UK	Tech	Engineering
1245	Christian	UK	Tech	Engineering
5375	Christian	UK	Tech	Engineering
3285	Christian	UK	Tech	Engineering
4525	Christian	UK	Tech	Engineering

n=4 → stillnot enough

Minimum sample size: n=40
(≈6 months recruiter output)

** the observation (**ID-0014**) will never appear in the n=40 sample population

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Find these reqs...	Christian	UK	Tech	Engineering	Entry
Find these reqs...	Christian	UK	Tech	Engineering	~~Entry~~
Find these reqs...	~~Christian~~	~~UK~~	~~Tech~~	~~Engineering~~	~~Entry~~
Find these reqs...	Christian	UK	~~Tech~~	~~Engineering~~	~~Entry~~
Find these reqs...	Christian	~~UK~~	~~Tech~~	~~Engineering~~	~~Entry~~
Find these reqs...	~~Christian~~	~~UK~~	~~Tech~~	~~Engineering~~	~~Entry~~

$n < 40 \rightarrow$ not enough

The **Context "Lens"**

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

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The **Context "Lens"**

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen	**not enough**		
time to fill			

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The **Context "Lens"**

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen	not enough		
time to fill			

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The **Context "Lens"**

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The **Context "Lens"**

	Recruiter	Org	Company
days to first screen			

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The **Context "Lens"**

days to first screen

Recruiter	Org	Company
• recruiter • • country • job_group • job_family • job_level	• country • job_group • job_family • job_level	• job_group • job_family • job_level

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Org Context “Lens”

Find these reqs...	UK	Tech	Engineering	Entry	
Find these reqs...	UK	Tech	Engineering	~~Entry~~	
Find these reqs...	UK	Tech	~~Engineering~~	~~Entry~~	n=40
Find these reqs...	UK	~~Tech~~	~~Engineering~~	~~Entry~~	n=40
Find these reqs...	~~UK~~	~~Tech~~	~~Engineering~~	~~Entry~~	n=40
Find these reqs...	~~UK~~	~~Tech~~	~~Engineering~~	~~Entry~~	n=40

The **Context "Lens"**

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen	not enough	**4** Samples Found	
time to fill			

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The **Context "Lens"**

	Recruiter	Org	Company
applicants total	not enough	4 Samples Found	3 Samples Found
reviewed in x days	not enough	4 Samples Found	3 Samples Found
days to first screen	not enough	4 Samples Found	3 Samples Found
time to fill	not enough	4 Samples Found	3 Samples Found

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Context

Recruiter
"Too New" for Samples

Org
16
Samples

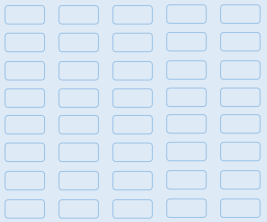
Company
12
Samples

1,120
Data Points

1
Requisition

28
Samples

40
Sample Size



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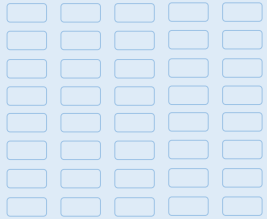
Q3
Requisition

vs.

Q2
Sample



Org Context "Lens"
UK | Tech | Engineering | Entry

1 Sample out of 28

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Statistics from **Context**

Q3 Requisition

ID-0014

Q2 Sample

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Q3 Requisition

ID-0014

Statistics from Context

Q2 Sample

Z-Score

Survival Range

25th to 75th percentile of the context sample

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Q3 Requisition

ID-0014

Q2 Sample

Metric	Z-Score	Sample Mean	Sample SD
15 days to first screen	-0.39 days to first screen	18 days to first screen	8 days to first screen

Survival Range

days to first screen

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Q3 Requisition

ID-0014

Q2 Sample

Metric

15
days to first screen

Z-Score

-0.39
days to first screen

Sample Mean

18
days to first screen

Sample SD

8
days to first screen

Survival Range

11 to 24 Days

days to first screen

1 Row of Data

Requisition ID	Context ID	Metric Name	Metric Value	Z-Score	Sample Mean	Sample SD	Survival Range
ID-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15 Days	-0.39	18	8	11 to 24 Days

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Q3 Requisition

ID-0014

Q2 Sample

Metric

15
days to first screen

Z-Score

-0.39
days to first screen

Sample Mean

18
days to first screen

Sample SD

8
days to first screen

Survival Range

11 to 24 Days

days to first screen

1 Row of Data

Requisition ID	Context ID	Metric Name	Metric Value	Z-Score	Sample Mean	Sample SD	Survival Range
ID-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15 Days	-0.39	18	8	11 to 24 Days

Org Context of "UK Tech Engineering Entry"resolves into a single standardized row

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All Rows for Reqisition ID-0014

1 Sample
out of 28

ID	Context ID	Metric Name	Metric Value	Z-Score	Sample Mean	Sample SD	Survival Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	89	1.11	50	36	19 to 77
RIQ-0014	org_UK_Tech_Engineering	applicants_total	89	0.52	67	42	27 to 99
RIQ-0014	org_UK_Tech	applicants_total	89	0.52	67	42	27 to 99
RIQ-0014	org_UK	applicants_total	89	0.48	69	41	38 to 94
RIQ-0014	company_Tech_Engineering_Entry	applicants_total	89	1.15	52	32	28 to 69
RIQ-0014	company_Tech_Engineering	applicants_total	89	0.58	64	43	25 to 96
RIQ-0014	company_Tech	applicants_total	89	0.58	64	43	25 to 96
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15	-0.39	18	8	11 to 24
RIQ-0014	org_UK_Tech_Engineering	days_to_first_screen	15	-0.33	18	9	9 to 25
RIQ-0014	org_UK_Tech	days_to_first_screen	15	-0.33	18	9	9 to 25
RIQ-0014	org_UK	days_to_first_screen	15	-0.22	17	7	10 to 22
RIQ-0014	company_Tech_Engineering_Entry	days_to_first_screen	15	-0.07	16	8	7 to 21
RIQ-0014	company_Tech_Engineering	days_to_first_screen	15	-0.11	16	8	8 to 23
RIQ-0014	company_Tech	days_to_first_screen	15	-0.11	16	8	8 to 23
RIQ-0014	org_UK_Tech_Engineering_Entry	percent_reviewed_in_x_days	62%	0.44	47%	34%	15% to 80%
RIQ-0014	org_UK_Tech_Engineering	percent_reviewed_in_x_days	62%	0.43	46%	37%	8% to 81%
RIQ-0014	org_UK_Tech	percent_reviewed_in_x_days	62%	0.43	46%	37%	8% to 81%
RIQ-0014	org_UK	percent_reviewed_in_x_days	62%	0.55	42%	35%	14% to 68%
RIQ-0014	company_Tech_Engineering_Entry	percent_reviewed_in_x_days	62%	0.91	33%	31%	8% to 53%
RIQ-0014	company_Tech_Engineering	percent_reviewed_in_x_days	62%	0.44	45%	38%	8% to 76%
RIQ-0014	company_Tech	percent_reviewed_in_x_days	62%	0.44	45%	38%	8% to 76%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	62	-0.92	68	7	62 to 72
RIQ-0014	org_UK_Tech_Engineering	time_to_fill	62	0.91	48	16	35 to 49
RIQ-0014	company_Tech_Engineering_Entry	time_to_fill	62	-1.48	74	8	68 to 79
RIQ-0014	org_UK_Tech	time_to_fill	62	0.91	48	16	35 to 49
RIQ-0014	company_Tech_Engineering	time_to_fill	62	0.16	60	15	47 to 70
RIQ-0014	org_UK	time_to_fill	62	0.2	58	23	39 to 76
RIQ-0014	company_Tech	time_to_fill	62	0.16	60	15	47 to 70

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Normal	Watch	Monitor	Inspect
days_to_first_screen reviewed_in_x_days	time_to_fill	applicants_total	

Requisition ID	Context ID	Metric Name	Metric Value	Z-Score
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	89	1.11
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15	0.39
RIQ-0014	org_UK_Tech_Engineering_Entry	reviewed_in_x_days	62%	0.44
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	62	0.92

How did Requisition ID-0014 compare to *Org Context of "UK Tech Engineering Entry"*

- Normal (|z| < 0.85)
- Watch (|z| < 1.1)
- Monitor (|z| < 1.96)
- Inspect (|z| > 1.96)

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All Rows for Reqisition ID-0014

ID	Context ID	Metric Name	Signal	Z-Score
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	Monitor	1.11
RIQ-0014	org_UK_Tech_Engineering	applicants_total	Normal	0.52
RIQ-0014	org_UK_Tech	applicants_total	Normal	0.52
RIQ-0014	org_UK	applicants_total	Normal	0.48
RIQ-0014	company_Tech_Engineering_Entry	applicants_total	Monitor	1.15
RIQ-0014	company_Tech_Engineering	applicants_total	Normal	0.58
RIQ-0014	company_Tech	applicants_total	Normal	0.58
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	Normal	-0.39
RIQ-0014	org_UK_Tech_Engineering	days_to_first_screen	Normal	-0.33
RIQ-0014	org_UK_Tech	days_to_first_screen	Normal	-0.33
RIQ-0014	org_UK	days_to_first_screen	Normal	-0.22
RIQ-0014	company_Tech_Engineering_Entry	days_to_first_screen	Normal	-0.07
RIQ-0014	company_Tech_Engineering	days_to_first_screen	Normal	-0.11
RIQ-0014	company_Tech	days_to_first_screen	Normal	-0.11
RIQ-0014	org_UK_Tech_Engineering_Entry	percent_reviewed_in_x_days	Normal	0.44
RIQ-0014	org_UK_Tech_Engineering	percent_reviewed_in_x_days	Normal	0.43
RIQ-0014	org_UK_Tech	percent_reviewed_in_x_days	Normal	0.43
RIQ-0014	org_UK	percent_reviewed_in_x_days	Normal	0.55
RIQ-0014	company_Tech_Engineering_Entry	percent_reviewed_in_x_days	Watch	0.91
RIQ-0014	company_Tech_Engineering	percent_reviewed_in_x_days	Normal	0.44
RIQ-0014	company_Tech	percent_reviewed_in_x_days	Normal	0.44
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	Watch	-0.92
RIQ-0014	org_UK_Tech_Engineering	time_to_fill	Watch	0.91
RIQ-0014	company_Tech_Engineering_Entry	time_to_fill	Monitor	-1.48
RIQ-0014	org_UK_Tech	time_to_fill	Watch	0.91
RIQ-0014	company_Tech_Engineering	time_to_fill	Normal	0.16
RIQ-0014	org_UK	time_to_fill	Normal	0.2
RIQ-0014	company_Tech	time_to_fill	Normal	0.16

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			Metric Value		Survival Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	89	?	19 to 77
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15		11 to 24
RIQ-0014	org_UK_Tech_Engineering_Entry	reviewed_in_x_days	62%		15% to 80%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	62		62 to 72

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			Metric Value	Signal	Survival Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	89	Out of Survival Range	19 to 77
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15	Within Survival Range	11 to 24
RIQ-0014	org_UK_Tech_Engineering_Entry	reviewed_in_x_days	62%	Within Survival Range	15% to 80%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	62	Within Survival Range	62 to 72

How did Requisition ID-0014 compare to *Org Context of "UK Tech Engineering Entry"*

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All Rows for Reqisition ID-0014

ID	Context ID	Metric Name	Signal	Metric Value	Survival Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	Out of Range	89	19 to 77
RIQ-0014	org_UK_Tech_Engineering	applicants_total	Within Range	89	27 to 99
RIQ-0014	org_UK_Tech	applicants_total	Within Range	89	27 to 99
RIQ-0014	org_UK	applicants_total	Within Range	89	38 to 94
RIQ-0014	company_Tech_Engineering_Entry	applicants_total	Out of Range	89	28 to 69
RIQ-0014	company_Tech_Engineering	applicants_total	Within Range	89	25 to 96
RIQ-0014	company_Tech	applicants_total	Within Range	89	25 to 96
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	Within Range	15	11 to 24
RIQ-0014	org_UK_Tech_Engineering	days_to_first_screen	Within Range	15	9 to 25
RIQ-0014	org_UK_Tech	days_to_first_screen	Within Range	15	9 to 25
RIQ-0014	org_UK	days_to_first_screen	Within Range	15	10 to 22
RIQ-0014	company_Tech_Engineering_Entry	days_to_first_screen	Within Range	15	7 to 21
RIQ-0014	company_Tech_Engineering	days_to_first_screen	Within Range	15	8 to 23
RIQ-0014	company_Tech	days_to_first_screen	Within Range	15	8 to 23
RIQ-0014	org_UK_Tech_Engineering_Entry	percent_reviewed_in_x_days	Within Range	62%	15% to 80%
RIQ-0014	org_UK_Tech_Engineering	percent_reviewed_in_x_days	Within Range	62%	8% to 81%
RIQ-0014	org_UK_Tech	percent_reviewed_in_x_days	Within Range	62%	8% to 81%
RIQ-0014	org_UK	percent_reviewed_in_x_days	Within Range	62%	14% to 68%
RIQ-0014	company_Tech_Engineering_Entry	percent_reviewed_in_x_days	Out of Range	62%	8% to 53%
RIQ-0014	company_Tech_Engineering	percent_reviewed_in_x_days	Within Range	62%	8% to 76%
RIQ-0014	company_Tech	percent_reviewed_in_x_days	Within Range	62%	8% to 76%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	Within Range	62	62 to 72
RIQ-0014	org_UK_Tech_Engineering	time_to_fill	Out of Range	62	35 to 49
RIQ-0014	company_Tech_Engineering_Entry	time_to_fill	Out of Range	62	68 to 79
RIQ-0014	org_UK_Tech	time_to_fill	Out of Range	62	35 to 49
RIQ-0014	company_Tech_Engineering	time_to_fill	Within Range	62	47 to 70
RIQ-0014	org_UK	time_to_fill	Within Range	62	39 to 76
RIQ-0014	company_Tech	time_to_fill	Within Range	62	47 to 70

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			Metric Value		Target Goal Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	89	?	5 to 100
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15		0 to 14
RIQ-0014	org_UK_Tech_Engineering_Entry	reviewed_in_x_days	62%		75% to 100%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	62		15 to 121

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			Metric Value	Signal	Target Goal Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	89	Met	5 to 100
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	15	Missed	0 to 14
RIQ-0014	org_UK_Tech_Engineering_Entry	reviewed_in_x_days	62%	Missed	75% to 100%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	62	Met	15 to 90

How did Requisition ID-0014 compare to *Org Context of "UK Tech Engineering Entry"*

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All Rows for Reqisition ID-0014

ID	Context ID	Metric Name	Signal	Metric Value	Suvival Range
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	Met	89	5 to 100
RIQ-0014	org_UK_Tech_Engineering	applicants_total	Met	89	5 to 100
RIQ-0014	org_UK_Tech	applicants_total	Met	89	5 to 100
RIQ-0014	org_UK	applicants_total	Met	89	5 to 100
RIQ-0014	company_Tech_Engineering_Entry	applicants_total	Met	89	5 to 100
RIQ-0014	company_Tech_Engineering	applicants_total	Met	89	5 to 100
RIQ-0014	company_Tech	applicants_total	Met	89	5 to 100
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	org_UK_Tech_Engineering	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	org_UK_Tech	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	org_UK	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	company_Tech_Engineering_Entry	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	company_Tech_Engineering	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	company_Tech	days_to_first_screen	Missed	15	0 to 14
RIQ-0014	org_UK_Tech_Engineering_Entry	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	org_UK_Tech_Engineering	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	org_UK_Tech	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	org_UK	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	company_Tech_Engineering_Entry	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	company_Tech_Engineering	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	company_Tech	percent_reviewed_in_x_days	Missed	62%	75% to 100%
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	Met	62	15 to 90
RIQ-0014	org_UK_Tech_Engineering	time_to_fill	Met	62	15 to 90
RIQ-0014	company_Tech_Engineering_Entry	time_to_fill	Met	62	15 to 90
RIQ-0014	org_UK_Tech	time_to_fill	Met	62	15 to 90
RIQ-0014	company_Tech_Engineering	time_to_fill	Met	62	15 to 90
RIQ-0014	org_UK	time_to_fill	Met	62	15 to 90
RIQ-0014	company_Tech	time_to_fill	Met	62	15 to 90

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All Rows for Reqisition ID-0014

ID	Context ID	Metric Name	Signal (Z-Score)	Signal (Survival Range)	Signal (Target Goal Range)
RIQ-0014	org_UK_Tech_Engineering_Entry	applicants_total	Monitor	Out of Range	Met
RIQ-0014	org_UK_Tech_Engineering	applicants_total	Normal	Within Range	Met
RIQ-0014	org_UK_Tech	applicants_total	Normal	Within Range	Met
RIQ-0014	org_UK	applicants_total	Normal	Within Range	Met
RIQ-0014	company_Tech_Engineering_Entry	applicants_total	Monitor	Out of Range	Met
RIQ-0014	company_Tech_Engineering	applicants_total	Normal	Within Range	Met
RIQ-0014	company_Tech	applicants_total	Normal	Within Range	Met
RIQ-0014	org_UK_Tech_Engineering_Entry	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	org_UK_Tech_Engineering	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	org_UK_Tech	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	org_UK	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	company_Tech_Engineering_Entry	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	company_Tech_Engineering	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	company_Tech	days_to_first_screen	Normal	Within Range	Missed
RIQ-0014	org_UK_Tech_Engineering_Entry	percent_reviewed_in_x_days	Normal	Within Range	Missed
RIQ-0014	org_UK_Tech_Engineering	percent_reviewed_in_x_days	Normal	Within Range	Missed
RIQ-0014	org_UK_Tech	percent_reviewed_in_x_days	Normal	Within Range	Missed
RIQ-0014	org_UK	percent_reviewed_in_x_days	Normal	Within Range	Missed
RIQ-0014	company_Tech_Engineering_Entry	percent_reviewed_in_x_days	Watch	Out of Range	Missed
RIQ-0014	company_Tech_Engineering	percent_reviewed_in_x_days	Normal	Within Range	Missed
RIQ-0014	company_Tech	percent_reviewed_in_x_days	Normal	Within Range	Missed
RIQ-0014	org_UK_Tech_Engineering_Entry	time_to_fill	Watch	Within Range	Met
RIQ-0014	org_UK_Tech_Engineering	time_to_fill	Watch	Out of Range	Met
RIQ-0014	company_Tech_Engineering_Entry	time_to_fill	Monitor	Out of Range	Met
RIQ-0014	org_UK_Tech	time_to_fill	Watch	Out of Range	Met
RIQ-0014	company_Tech_Engineering	time_to_fill	Normal	Within Range	Met
RIQ-0014	org_UK	time_to_fill	Normal	Within Range	Met
RIQ-0014	company_Tech	time_to_fill	Normal	Within Range	Met

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“I want to **compare Q3** to **Q2**”

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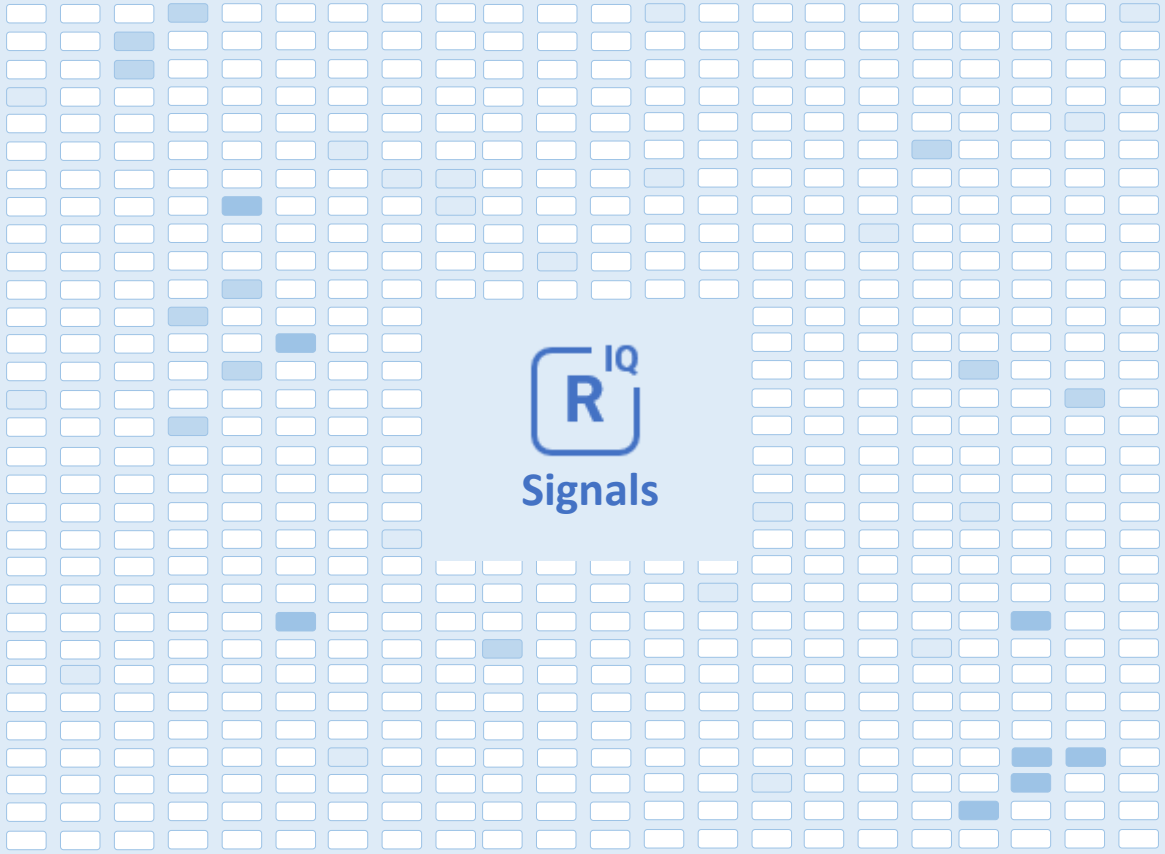
2,336 Q3 Requisitions

3,200,000 Context Comparisons



(78,640 Context IDs × 40 samples each, derived from 4,420 Q2 requisitions)

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This is the **Python Plumbing** underneath **RequisitionIQ**

```python
def add_signal_positive_z_scores_to_df(df, target_column):
    df[f"{target_column}_positive_only"] = np.abs(df[target_column])
    return df

def add_signal_floor_ceiling_to_df(df, schema, metric_config):
    df["metric_config_goal_floor"] = df["lineage_target_column"].map(
        {metric: schema["columns"][metric]["floor"] for metric in metric_config}
    )
    df["metric_config_goal_ceiling"] = df["lineage_target_column"].map(
        {metric: schema["columns"][metric]["ceiling"] for metric in metric_config}
    )
    df["metric_config_goal_range"] = df["metric_config_goal_ceiling"] -
df["metric_config_goal_floor"]
    df["metric_config_goal_mean"] = (df["metric_config_goal_floor"] +
df["metric_config_goal_ceiling"]) / 2
    df["metric_config_goal_std"] = df["metric_config_goal_range"] / 4
    return df

def add_signal_direction_to_df(df, schema, metric_config):
    df["metric_config_direction"] = df["lineage_target_column"].map(
    {metric: schema["columns"][metric]["direction"] for metric in metric_config}
    )
    return df
```

```python
def new_lineage_df_req_iq_cluster(
    df, target_column, req_iq_cluster_list_x, unique_id,
    top_n=None, sort_column=None, ascending_equals=None):
    concat_req_iq_cluster_df = {}
    for (cluster_name, cluster_group, cols, level) in req_iq_cluster_list_x:
        dict_name = f"{cluster_group}_{cols}"
        tmp_df = efunc.new_group_by_top_n_columns_df(
            df, cols, sort_column, top_n, ascending_equals, unique_id)
        tmp_df = efunc.add_lineage_columns(
            tmp_df, cluster_name, cluster_group, cols,
            level, sort_column, top_n, target_column)
        concat_req_iq_cluster_df[dict_name] = tmp_df
    return pd.concat(concat_req_iq_cluster_df)

def new_lineage_df_top_n(
    df, metrics, req_iq_cluster_list_x, unique_id, schema,
    sort_column=None, ascending_equals=None, top_n=None):
    concat_top_n_df = {}
    for metric in metrics:
        dict_name = f"{metric}_top_n_df"
        top_n_df = new_lineage_df_req_iq_cluster(
            df, metric, req_iq_cluster_list_x,
            unique_id, top_n, sort_column, ascending_equals)
        concat_top_n_df[dict_name] = efunc.new_mean_std_df(
```

```python
def new_mean_std_df(df, target_column, lineag
    if "mean" not in df.columns:
        df["mean"] = df.groupby(lineage_uniqu
    if "std" not in df.columns:
        df["std"] = df.groupby(lineage_unique
    return df

def add_z_score_to_df(df, target_column,
                      mean_name="mean", std_n
    df[column_name] = np.where(
        df[std_name] == 0, 0,
        (df[target_column] - df[mean_name]) /
    return df

def new_lineage_df_group_by_cumulative_count_
        df, target_column, lineage_unique_id)
    assert df["lineage_top_n"].max() is not N
    top_n = df["lineage_top_n"].max()
    df = df.groupby(lineage_unique_id)[target
            .value_counts().reset_index(name="
    df = df.sort_values(
        by=[lineage_unique_id, target_column]
        ascending=[True, True]).reset_index(d
    df["cum_count"] = (
```

```python
    df["metric_config_z_score_watch"] = df["lineage_target_column"].map(
        {metric: schema["columns"][metric]["z_score_watch"] for metric in metrics}
    )
    df["metric_config_z_score_monitor"] = df["lineage_target_column"].map(
        {metric: schema["columns"][metric]["z_score_monitor"] for metric in metrics}
    )
    df["metric_config_z_score_inspect"] = df["lineage_target_column"].map(
        {metric: schema["columns"][metric]["z_score_inspect"] for metric in metrics}
    )
    return df

def add_bin_z_score(df, target_column):
    z = df[target_column]
    df[f"bin_{target_column}"] = np.select(
        [
            z <= df["metric_config_z_score_normal"],
            z <= df["metric_config_z_score_watch"],
            z <= df["metric_config_z_score_monitor"]
        ],
        [0, 1, 2],
        default=3
    )
    return df
```

```python
    for metric in metrics:
        dict_name = f"{metric}_survival_df"
        survival_df = df[df["lineage_target_column"] == metric]
        survival_df =
efunc.new_lineage_df_group_by_cumulative_count_running_percent(
            survival_df, metric, "lineage_unique_id")
        survival_df = efunc.classify_lineage_df_survival_quarter_bins(
            survival_df, "percent", metric)
        concat_survival_df[dict_name] = pd.pivot_table(
            survival_df, values=metric,
            index=["lineage_unique_id","lineage_target_column"],
            columns=["bin_survival"], aggfunc="max")

    return pd.concat(concat_survival_df)

def new_group_by_top_n_columns_df(df, on_columns, sort_column,
                                  top_n, ascending_equals, unique_id):

    df = df.sort_values(by=sort_column, ascending=ascending_equals)
    df["count"] = df.groupby(on_columns)[unique_id].transform("count")

    if top_n is not None:
        df = df[df["count"] >= top_n]
        df["cumulative_count"] = (
```

```python
        df, percent_column, lineage_target_co
    df["bin_survival"] = pd.cut(
        df[percent_column],
        bins=[0, .25, .5, .75, 1],
        labels=["bin_survival_25th",
                "bin_survival_50th",
                "bin_survival_75th",
                "bin_survival_100th"])
    df["lineage_target_column"] = lineage_tar
    return df
```

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you define the **inputs**

```
Segment 1 [location]
  Level 2 ["TERRITORY", "COUNTRY"]
  Level 1 ["TERRITORY"]
```

```
Segment 2 [deal]
  Level 2 ["PRODUCTLINE", "DEALSIZE"]
  Level 1 ["PRODUCTLINE"]
```

```
Segment 3 [customer]
  Level 1 ["CUSTOMERNAME"]
```

```
Segment 4 [product]
  Level 1 ["PRODUCTCODE"]
```

```
Segment 5 [status]
  Level 2 ["STATUS", "PRODUCTLINE"]
  Level 1 ["STATUS"]
```

QUANTITY

```
dtype: int64
goal: 30
direction: 1
floor: 10
ceiling: 100
z_score_normal: 0.85
z_score_watch: 1.15
z_score_monitor: 1.96
z_score_inspect: 2.5
survival_25th: 0
survival_50th: 0.5
survival_75th: 1
```

PRICE

SALES

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the **Python Plumbing** runs…

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
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days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
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	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
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applicants total			
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	Recruiter	Org	Company
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	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
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	Recruiter	Org	Company
applicants total			
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days to first screen			
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	Recruiter	Org	Company
applicants total			
reviewed in x days			
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	Recruiter	Org	Company
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reviewed in x days			
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	Recruiter	Org	Company
applicants total			
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	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
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	Recruiter	Org	Company
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	Recruiter	Org	Company
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	Recruiter	Org	Company
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	Recruiter	Org	Company
applicants total			
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time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

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if a sample doesn’t equal n we drop it.

statistical minimums ensure reliable comparisons

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
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days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
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	Recruiter	Org	Company
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reviewed in x days			
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	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
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	Recruiter	Org	Company
applicants total			
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days to first screen			
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	Recruiter	Org	Company
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	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

	Recruiter	Org	Company
applicants total			
reviewed in x days			
days to first screen			
time to fill			

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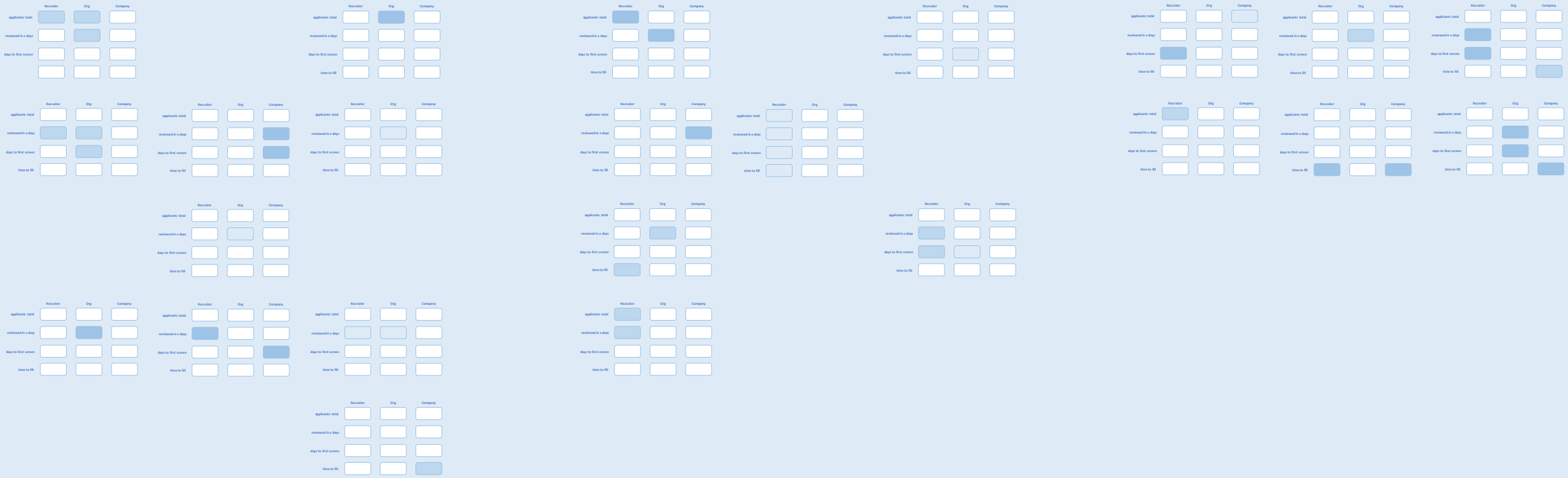
context is applied



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RequisitionIQ works on **sales** data…

ID	Context ID	Metric Name	Metric Value	Z-Score	Sample Mean	Sample SD	Survival Range
10208S12_1108	status_Shipped	QUANTITYORDERED	46	0.79	35	14	23 to 41
10208S12_1108	deal_Classic Cars_Large	SALES	8602	-0.05	8676	1560	7396 to 9240
10208S12_1108	deal_Classic Cars_Large	PRICEEACH	100	0.26	100	2	94 to 122
10208S12_1108	deal_Classic Cars_Large	QUANTITYORDERED	46	-0.52	53	14	46 to 59
10208S12_1108	deal_Classic Cars	SALES	8602	1.72	4274	2521	2575 to 5296
10208S12_1108	deal_Classic Cars	PRICEEACH	100	0.62	87	21	64 to 110
10208S12_1108	deal_Classic Cars	QUANTITYORDERED	46	0.57	38	14	28 to 46
10208S12_1108	location_EMEA_France	SALES	8602	1.87	3819	2561	1796 to 4692
10208S12_1108	location_EMEA_France	PRICEEACH	100	1.01	76	24	49 to 99
10208S12_1108	location_EMEA_France	QUANTITYORDERED	46	0.24	42	18	28 to 47
10208S12_1108	location_EMEA	SALES	8602	1.67	4045	2728	1781 to 5318
10208S12_1108	location_EMEA	PRICEEACH	100	0.72	85	21	61 to 130
10208S12_1108	location_EMEA	QUANTITYORDERED	46	0.72	36	14	26 to 44
10208S12_1108	status_Shipped_Classic Cars	SALES	8602	1.19	5016	3009	2896 to 6712
10208S12_1108	status_Shipped_Classic Cars	PRICEEACH	100	0.52	89	21	83 to 110
10208S12_1108	status_Shipped_Classic Cars	QUANTITYORDERED	46	0.43	39	16	27 to 47
10208S12_1108	status_Shipped	SALES	8602	2.03	3930	2303	1961 to 4764
10208S12_1108	status_Shipped	PRICEEACH	100	0.7	87	18	74 to 112

the exact same code processed both datasets.
I just changed the config file.

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It’s my **general-purpose** descriptive statistics engine

ID	Context ID	Metric Name	Signal (Z-Score)	Signal (Survival Range)	Signal (Target Goal Range)
10208S12_1108	status_Shipped	QUANTITYORDERED	Normal	Out of Range	Met
10208S12_1108	deal_Classic Cars_Large	SALES	Normal	Within Range	Met
10208S12_1108	deal_Classic Cars_Large	PRICEEACH	Normal	Within Range	Missed
10208S12_1108	deal_Classic Cars_Large	QUANTITYORDERED	Normal	Within Range	Met
10208S12_1108	deal_Classic Cars	SALES	Monitor	Out of Range	Met
10208S12_1108	deal_Classic Cars	PRICEEACH	Normal	Within Range	Missed
10208S12_1108	deal_Classic Cars	QUANTITYORDERED	Normal	Within Range	Met
10208S12_1108	location_EMEA_France	SALES	Monitor	Out of Range	Met
10208S12_1108	location_EMEA_France	PRICEEACH	Watch	Out of Range	Missed
10208S12_1108	location_EMEA_France	QUANTITYORDERED	Normal	Within Range	Met
10208S12_1108	location_EMEA	SALES	Monitor	Out of Range	Met
10208S12_1108	location_EMEA	PRICEEACH	Normal	Within Range	Missed
10208S12_1108	location_EMEA	QUANTITYORDERED	Normal	Out of Range	Met
10208S12_1108	status_Shipped_Classic Cars	SALES	Monitor	Out of Range	Met
10208S12_1108	status_Shipped_Classic Cars	PRICEEACH	Normal	Within Range	Missed
10208S12_1108	status_Shipped_Classic Cars	QUANTITYORDERED	Normal	Within Range	Met
10208S12_1108	status_Shipped	SALES	Inspect	Out of Range	Met
10208S12_1108	status_Shipped	PRICEEACH	Normal	Within Range	Missed

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The **RequisitionIQ** Framework

Level 4: Advanced Intelligence
Learn from requisition reality

- *Supervised Learning*
- *Unsupervised Learning*
- *Deep Learning*

Level 3: Pipeline Health
You can't manage requisitions when you can't monitor

- *Decision Analytics*
- *Classification*
- *Clustering*
- *Big Data Analytics*

Level 2: Context Intelligence
Predict the outcome of requisitions

- *Advanced Statistics*

Level 1: Historic Context
Define normal filled requisitions

Sampling	Context	Signals
Z-Scores	Survival	Target Goals

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R IQ